

AMRITA VISHWA VIDYAPEETHAM
AMARAVATI CAMPUS

Alumni Portal

Feature Documentation & Deployment Proposal

Prepared by	Student Development Team — Amrita Amaravati
Category	Internal Technical Proposal
Status	Development Complete — Awaiting Deployment
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1. Executive Summary

Amrita Vishwa Vidyapeetham operates an official alumni network portal at <https://alumni.amrita.edu/>, which serves alumni across all campuses. However, the existing portal provides only basic connectivity features and lacks a comprehensive, structured alumni data management system specifically tailored to the needs of the **Amaravati campus**. Key capabilities such as detailed career tracking, multi-step alumni onboarding, engagement preference management, mentorship and recruitment support features, and a powerful administrative filtering dashboard are entirely absent from the existing platform.

Recognising this gap, the student development team at Amrita Vishwa Vidyapeetham, Amaravati Campus has independently designed, architected, and fully developed a production-grade **Amrita Amaravati Alumni Portal** from scratch. The application is feature-complete and has undergone development and internal testing. The portal is now ready for deployment and requires only three resources from the university administration: a dedicated domain, a server environment, and cloud storage access.

Portal Status	Development Complete — Ready to Deploy
Existing Amrita Portal	https://alumni.amrita.edu/ (campus-agnostic, limited features)
This Portal	Amaravati-specific, feature-rich, production-grade
Resources Needed	Domain + Server + Storage Access
Built By	Amrita Vishwa Vidyapeetham, Amaravati — Student Development Team

2. Technology Stack

The portal is built on a modern, scalable, and production-proven stack:

Layer	Technology	Purpose
Frontend	Next.js 14 (App Router) + Tailwind CSS	Responsive UI, SSR, routing
Backend	FastAPI (Python)	REST API, async, auto-docs
Database	PostgreSQL (SQLAlchemy + Alembic)	Structured data, ACID, migrations
Auth	Firebase Authentication	Email/Password + Google OAuth
OTP / Email	Gmail SMTP (App Password) + Redis	Free email OTP, rate limiting
Storage	Firebase Storage	Photos, resumes, documents
Real-time	Firestore	Notifications, event updates
Cache	Redis	OTP TTL, rate limiting, sessions
Security	bcrypt + JWT + HTTPS + CORS + Pydantic	End-to-end security hardening

3. Complete Feature List

3.1 Account Creation & Authentication

- User signup with Full Name, Email, Password (strength-validated)
- Email OTP verification — 6-digit code sent via Gmail SMTP (free, no third-party SMS cost)
- OTP security: 5-minute expiry (Redis TTL), max 3 wrong attempts, 15-minute lockout, max 5 resends/hour

- Individual OTP input boxes with auto-focus and auto-submit on 6th digit
- 60-second countdown timer before resend is allowed
- Google OAuth login (Firebase one-click sign-in)
- Forgot password via Firebase email reset link
- Firebase custom token exchange — secure session management
- Remember me / persistent session support

3.2 Alumni Dashboard

- Post-login personalised dashboard with greeting
- Registration completion banner for incomplete profiles
- Profile summary card with quick stats
- Quick action buttons: View Profile, Edit Profile, Update Career Info

3.3 Multi-Step Alumni Registration Form (9 Steps)

The registration is structured as a guided, step-by-step application form. Progress is auto-saved after each step so alumni can close the browser and resume later. A sticky progress bar shows the current step and step name throughout.

Step 1 — Basic Personal Information

- Full Name (pre-filled from account)
- Gender
- Date of Birth
- Profile Photograph (drag-and-drop upload with crop preview, stored in Firebase Storage)
- Student ID / Roll Number
- Nationality

Step 2 — Contact Information

- Personal Mobile Number
- WhatsApp Number (same-as-mobile checkbox)
- Personal Email (pre-filled)
- Alternate Email ID
- LinkedIn Profile URL (URL format validated)
- Current Address
- Permanent Address (same-as-current checkbox)
- City, State, Country, PIN Code

Step 3 — Academic Information

- University / College Name
- Campus
- Department (searchable dropdown + Other)
- Program / Course
- Specialization
- Degree (B.Tech / M.Tech / MBA / PhD / B.Sc / M.Sc / Other)
- Batch / Passing Year
- Admission Year

Step 4 — Employment / Career Information

- Current Employment Status (Employed / Higher Studies / Entrepreneur / Government Services / Preparing for Exams)
- Current Company Name, Designation, Industry Sector
- Employment Type (Full-time / Part-time / Contract / Freelance / Remote)
- Work Location (City, Country)
- Annual CTC Range (optional)
- Previous Companies — dynamic add/remove list (Company, Designation, From/To Year)
- Years of Experience
- Skills / Technologies (tag input)

Step 5 — Higher Education Information

- University Name, Country
- Degree Pursued, Course / Specialization
- Admission Year, Graduation Year
- "Currently Pursuing" toggle

Step 6 — Entrepreneurship Details

- Startup Name, Domain / Industry
- Website URL
- Funding Status (Bootstrapped / Angel / Seed / Series A+ / Not Applicable)
- Team Size
- Incubation Details

Step 7 — Alumni Engagement Preferences

- Interested in Mentoring Students?
- Interested in Guest Lectures?
- Interested in Recruitment Support?
- Interested in Internship Support?
- Interested in Donations / CSR?
- Willing to Participate in Alumni Chapters?
- Preferred Communication Mode (Email / Phone / WhatsApp / LinkedIn)
- Available for Campus Events?

Step 8 — Placement & Networking Support

- Can Refer Students?
- Hiring Capacity (number of hires this year)
- Internship Opportunities Available?
- Industry Contacts
- Recruitment Areas / Domains (tag input)

Step 9 — Social Profiles, Documents & Review

- LinkedIn URL, GitHub Profile URL, Portfolio Website URL
- Resume Upload (PDF only, max 5MB, Firebase Storage)
- Optional additional document upload
- Full review summary — expandable per section with Edit links
- Accuracy confirmation checkbox before final submit
- Confetti animation on successful registration completion

3.4 Alumni Profile Page

- Tabbed profile layout: Overview | Career | Education | Engagement | Social
- Profile photo prominently shown with name, batch, department, degree
- Career: current role card + previous companies timeline
- Education: academic and higher education displayed as clean cards
- Engagement: mentor / lecture / referral availability as status badges
- Social: LinkedIn, GitHub, Portfolio as styled link buttons
- Resume download button
- Edit Profile button — opens pre-filled multi-step form
- Privacy: alumni can only access their own profile (403 on others)

3.5 Admin Panel (/admin)

The admin panel is completely separate from the alumni authentication system. It uses its own username + bcrypt-hashed password stored securely in environment variables, with an httpOnly JWT session cookie (8-hour expiry). No alumni Firebase token can ever grant admin access.

- Summary stat cards: Total Alumni, Fully Registered, Employed, Higher Studies, Entrepreneurs, Mentors Available, Can Refer Students
- Full alumni data table: Photo thumbnail, Name, Batch, Dept, Degree, Company, Designation, Location, Status, Registered On
- Pagination (25 per page) with total count display
- Search bar: simultaneous search across name, email, company, student ID, designation

Advanced Filters (all combinable with AND logic):

- Batch Year (multi-select)
- Department (multi-select)
- Degree (multi-select)
- Employment Status (multi-select)
- Company Name (partial text match)
- Designation (partial text match)
- Industry Sector (multi-select)
- Employment Type (multi-select)
- Work City (text) / Country (dropdown)
- Skills / Technologies (tag search — alumni with ALL selected skills)
- Mentoring Available (All / Yes / No)
- Guest Lecture Available (All / Yes / No)
- Can Refer Students (All / Yes / No)
- Internship Opportunities (All / Yes / No)
- Graduation Year range (From → To)

- Registration Date range (date picker)
- Active filter count shown as badge on filter panel toggle button
- URL query params update on filter apply — filters survive page refresh
- Sortable table columns (click header to sort)
- Export to CSV — exports ALL current filtered results (not just visible page)
- Alumni detail modal — full profile view including all optional fields
- Admin logout — clears httpOnly cookie

4. Complete Data Fields Captured

The portal captures the following structured data for every registered alumni:

Basic Personal Information	Full Name Gender Date of Birth Profile Photograph Student ID / Roll Number Nationality
Contact Information	Personal Mobile Number WhatsApp Number Personal Email ID Alternate Email ID LinkedIn Profile URL Current Address Permanent Address City, State, Country, PIN Code
Academic Information	University / College Name Campus Department Program / Course Specialization Degree Batch / Passing Year Admission Year
Employment / Career	Current Employment Status Current Company Name Designation Industry Sector Employment Type Work Location Annual CTC Range (optional) Previous Companies (history) Years of Experience Skills / Technologies
Higher Education	University Name Country Degree Pursued Specialization Admission Year Graduation Year / Ongoing
Entrepreneurship	Startup Name Domain / Industry Website Funding Status Team Size Incubation Details
Engagement Preferences	Mentoring, Guest Lectures, Recruitment Support Internship Support, Donations/CSR, Alumni Chapters Preferred Communication Mode Availability for Campus Events
Placement & Networking	Can Refer Students Hiring Capacity Internship Opportunities Industry Contacts Recruitment Areas
Social & Professional	LinkedIn, GitHub, Portfolio URLs Resume (PDF) Additional Document
Activity Tracking	Last Interaction Date Events Attended Donations Made Talks Delivered Mentorship Sessions Awards / Achievements Alumni Membership Status

Note: Aadhaar number, Passport number, and Employee ID fields have been intentionally excluded from the portal to protect alumni privacy and comply with data minimisation principles under the DPDP Act, 2023.

5. Security Architecture

Given that this portal will be used by a large number of alumni of a major university and contains personal and professional data, the following security measures have been implemented:

Firebase ID Token Verification	Every protected alumni API call verifies the Firebase ID token on the backend using firebase-admin SDK.
Email OTP with Redis TTL	OTPs are 6-digit cryptographically generated codes stored only in Redis with a 5-minute TTL — never in the database.
OTP Brute-Force Protection	Max 3 failed attempts triggers a 15-minute email lockout. Max 5 resend requests per hour per email.
Admin Credential Security	Admin username and bcrypt-hashed password stored in environment variables only. Never stored in the database.

Admin Session Isolation	Admin uses httpOnly + Secure + SameSite=Strict JWT cookies. Completely separate from Firebase alumni auth — no cross-access possible.
CORS Policy	Only the configured frontend domain is whitelisted. All other origins are rejected.
Rate Limiting	100 API requests per minute per IP across all routes (slowapi).
File Upload Validation	MIME type checked (not just extension), max 5MB per file, executable files rejected, filenames sanitised.
Input Validation	All inputs validated via Pydantic schemas. No raw user input passed to queries.
SQL Injection Prevention	SQLAlchemy ORM used exclusively — zero raw SQL strings in the codebase.
Row-Level Data Isolation	Alumni can only read and write their own records. User ID verified on every database query.
Security Headers	X-Content-Type-Options, X-Frame-Options: DENY, Referrer-Policy, Permissions-Policy on all responses.
HTTPS Enforcement	Middleware redirects all HTTP requests to HTTPS.
Soft Deletes Only	Alumni records are never hard-deleted — only flagged as inactive. Ensures data integrity and audit trail.
Password Storage	Alumni passwords are never stored in PostgreSQL — fully managed by Firebase Authentication.

6. Database Schema Overview

The portal uses a fully normalised PostgreSQL schema with 12 tables, proper foreign key constraints, UUID primary keys, and GIN indexes for array-based searches (skills, recruitment areas).

Table Name	Key Fields	Purpose
users	id, firebase_uid, full_name, email, mobile, is_email_verified	Core account record linked to Firebase
alumni_profile	user_id, student_id, gender, dob, photo_url, nationality, registration_complete, current_step	Master alumni record
alumni_contact	mobile, whatsapp, alternate_email, linkedin_url, current_address, permanent_address, city, state, country, pin_code	All contact information
alumni_academic	university_name, campus, department, program, specialization, degree, batch_year, admission_year	Academic background
alumni_career	employment_status, current_company, designation, industry_sector, employment_type, work_city, work_country, ctc_range, years_of_experience, skills[]	Current career details
alumni_career_history	company_name, designation, from_year, to_year	Previous employment (multiple rows per alumni)
alumni_higher_education	university_name, country, degree, specialization, admission_year, graduation_year, is_ongoing	PG / abroad studies
alumni_entrepreneurship	startup_name, domain, website, funding_status, team_size, incubation_details	Startup information

alumni_engagement	mentoring, guest_lectures, recruitment_support, internship_support, donations_csr, alumni_chapters, preferred_communication, available_for_events	Engagement preferences
alumni_networking	can_refer, hiring_capacity, internship_opportunities, industry_contacts, recruitment_areas[]	Networking & placement support
alumni_social	linkedin, github, portfolio, resume_url, other_doc_url	Social profiles & documents
alumni_activity	last_interaction, events_attended[], donations_made, talks_delivered, mentorship_sessions, awards_achievements, membership_status	Activity & contributions

7. Comparison with Existing Amrita Alumni Portal

The official Amrita alumni network is accessible at <https://alumni.amrita.edu/>. While it serves the broader Amrita network, it does not provide the granular, campus-specific features required for effective alumni engagement at the Amaravati campus. The table below highlights the key differences:

Feature	alumni.amrita.edu (Existing Portal)	Amaravati Alumni Portal (This Application)
Campus-specific registration	✗ Not available	✓ Fully built for Amaravati
Multi-step guided registration form	✗ Not available	✓ 9-step structured form
Email OTP verification	✗ Not available	✓ Implemented (free, Gmail SMTP)
Career history tracking	✗ Not available	✓ Dynamic multi-entry career history
Entrepreneurship details	✗ Not available	✓ Full startup profile section
Higher education tracking	✗ Not available	✓ PG / abroad studies section
Alumni engagement preferences	✗ Not available	✓ Mentoring, lectures, CSR, chapters
Placement & referral support	✗ Not available	✓ Referral, hiring capacity, internships
Resume / document upload	✗ Not available	✓ Firebase Storage (PDF, 5MB)
Skill tag search	✗ Not available	✓ GIN-indexed PostgreSQL array search
Admin panel with advanced filters	✗ Not available	✓ 16+ combinable filters
CSV export of filtered alumni	✗ Not available	✓ Full filtered export
Activity tracking (events, talks)	✗ Not available	✓ Full activity log per alumni
Google OAuth login	✗ Not available	✓ Firebase Google sign-in
Production security hardening	Unknown	✓ 15 security measures implemented
Admin isolation (separate auth)	Unknown	✓ Completely separate from alumni auth

8. Deployment Requirements

The application is fully developed and internally tested. The development team at Amrita Vishwa Vidyapeetham, Amaravati Campus requests the following three resources from the university administration to proceed with deployment:

1

Domain

A dedicated subdomain such as alumni.amaravati.amrita.edu or a new domain. This will be the public-facing URL of the portal for all registered alumni and the admin panel.

2

Server / Hosting Environment

A Linux-based server (Ubuntu 22.04+) or a cloud hosting environment (AWS EC2 / Azure App Service / Google Cloud Run) capable of running: FastAPI backend (Python), Next.js frontend (Node.js), PostgreSQL database, Redis instance. Minimum recommended: 2 vCPU, 4GB RAM, 20GB storage for the application and database. Can also use Railway, Render, or Vercel (frontend) + Supabase (managed PostgreSQL) for a low-cost deployment.

3

Cloud Storage Access

Firebase project access (Firebase Storage bucket) OR an S3-compatible object storage bucket for storing: alumni profile photographs, uploaded resumes (PDF), and any additional documents. Firebase Storage free tier (5GB) is sufficient for initial deployment and scales automatically on the Blaze plan.

No additional development work is required. The codebase is complete, version-controlled, and documented. Once the above three resources are provisioned, the team can deploy the portal within 24–48 hours.

9. Configuration & Environment Variables

The following environment variables must be configured at the time of deployment:

Variable	Description
DATABASE_URL	PostgreSQL connection string (asyncpg)
REDIS_URL	Redis connection string
FIREBASE_SERVICE_ACCOUNT_JSON	Base64-encoded Firebase Admin SDK service account JSON
FIREBASE_STORAGE_BUCKET	Firebase Storage bucket name
ADMIN_USERNAME	Admin panel login username
ADMIN_PASSWORD_HASH	bcrypt hash of admin password (generated via provided script)
ADMIN_JWT_SECRET	64-character random string for admin JWT signing
CORS_ORIGIN	Frontend domain (e.g. https://alumni.amaravati.amrita.edu)
SMTP_HOST / SMTP_PORT	Gmail SMTP server (smtp.gmail.com / 587)
SMTP_USER / SMTP_PASSWORD	Gmail address and App Password for OTP emails
NEXT_PUBLIC_FIREBASE_*	Firebase client-side config keys (API key, auth domain, project ID, etc.)
NEXT_PUBLIC_API_BASE_URL	Backend API base URL

10. UI / UX Design Standards

- Professional university-grade design — deep navy, white, and gold color palette
- Fully responsive — works seamlessly on mobile phones, tablets, and desktops
- Toast notifications (react-hot-toast) for all success, error, and info states
- Loading spinners on all async operations
- Skeleton loaders on admin table and profile page while data loads

- Inline form validation — per-field error messages shown as user types
- Password strength meter on signup page
- Drag-and-drop file upload with preview for profile photo and resume
- Unsaved changes browser warning if user navigates away mid-registration
- Confetti animation on successful registration completion
- Admin filter panel shows active filter count as a number badge
- Admin table columns are sortable by clicking the column header

11. Deliverables Summary

Complete Next.js 14 Frontend	All pages, components, and routes fully implemented and tested
Complete FastAPI Backend	All API routes, models, schemas, and utilities
PostgreSQL Migrations (Alembic)	All 12 tables with indexes and constraints
Email OTP System	Gmail SMTP integration, Redis TTL, brute-force protection
Firebase Integration	Auth, Storage, and Firestore configuration
Admin Panel	Full filtering, search, CSV export, detail modal
Environment Variable Templates	backend/.env.example and frontend/.env.local.example
Admin Credential Setup Script	scripts/generate_admin_hash.py — one-command setup
Docker Compose (Local Dev)	postgres + redis + backend + frontend services
README.md	Step-by-step local setup and deployment instructions

12. Conclusion

The Amrita Amaravati Alumni Portal represents a significant advancement over the existing campus-agnostic portal at alumni.amrita.edu. It is purpose-built for the Amaravati campus, captures comprehensive alumni data across 10 structured categories, provides a powerful administrative interface for filtering and engaging alumni, and is built to production-grade security and performance standards.

The student development team has completed the entire application and is ready for immediate deployment. The only requirement from the university administration is provisioning of a **domain**, a **server environment**, and **cloud storage access**. No additional development cost or timeline is involved.

We respectfully request the university's support in providing these resources so that the Amaravati alumni community can benefit from a modern, secure, and feature-rich portal that reflects the academic excellence of Amrita Vishwa Vidyapeetham.

Prepared by	Institution	Date
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